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CSES Problem Set

Point Location Test

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a line that goes through the points $p_1 = (x_1, y_1)$ and $p_2 = (x_2, y_2)$. There is also a point $p_3 = (x_3, y_3)$.

Your task is to determine whether p_3 is located on the left or right side of the line or if it touches the line when we are looking from p_1 to p_2 .

Input

The first input line has an integer t : the number of tests.

After this, there are t lines that describe the tests. Each line has six integers: x_1, y_1, x_2, y_2, x_3 and y_3 .

Output

For each test, print "LEFT", "RIGHT" or "TOUCH".

Constraints

- $1 \leq t \leq 10^5$
- $-10^9 \leq x_1, y_1, x_2, y_2, x_3, y_3 \leq 10^9$
- $x_1 \neq x_2$ or $y_1 \neq y_2$

Example

Input:

```
3
1 1 5 3 2 3
1 1 5 3 4 1
1 1 5 3 3 2
```

Output:

```
LEFT
RIGHT
TOUCH
```

CSES Problem Set**Line Segment Intersection**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are two line segments: the first goes through the points (x_1, y_1) and (x_2, y_2) , and the second goes through the points (x_3, y_3) and (x_4, y_4) .

Your task is to determine if the line segments intersect, i.e., they have at least one common point.

Input

The first input line has an integer t : the number of tests.

After this, there are t lines that describe the tests. Each line has eight integers $x_1, y_1, x_2, y_2, x_3, y_3, x_4$ and y_4 .

Output

For each test, print "YES" if the line segments intersect and "NO" otherwise.

Constraints

- $1 \leq t \leq 10^5$
- $-10^9 \leq x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4 \leq 10^9$
- $(x_1, y_1) \neq (x_2, y_2)$
- $(x_3, y_3) \neq (x_4, y_4)$

Example

Input:

```
5
1 1 5 3 1 2 4 3
1 1 5 3 1 1 4 3
1 1 5 3 2 3 4 1
1 1 5 3 2 4 4 1
1 1 5 3 3 2 7 4
```

Output:

```
NO
YES
YES
YES
YES
```

CSES Problem Set

Polygon Area

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to calculate the area of a given polygon.

The polygon consists of n vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The vertices (x_i, y_i) and (x_{i+1}, y_{i+1}) are adjacent for $i = 1, 2, \dots, n - 1$, and the vertices (x_1, y_1) and (x_n, y_n) are also adjacent.

Input

The first input line has an integer n : the number of vertices.

After this, there are n lines that describe the vertices. The i th such line has two integers x_i and y_i .

You may assume that the polygon is simple, i.e., it does not intersect itself.

Output

Print one integer: $2a$ where the area of the polygon is a (this ensures that the result is an integer).

Constraints

- $3 \leq n \leq 1000$
- $-10^9 \leq x_i, y_i \leq 10^9$

Example

Input:

```
4
1 1
4 2
3 5
1 4
```

Output:

```
16
```

CSES Problem Set

Point in Polygon

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a polygon of n vertices and a list of m points. Your task is to determine for each point if it is inside, outside or on the boundary of the polygon.

The polygon consists of n vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The vertices (x_i, y_i) and (x_{i+1}, y_{i+1}) are adjacent for $i = 1, 2, \dots, n - 1$, and the vertices (x_1, y_1) and (x_n, y_n) are also adjacent.

Input

The first input line has two integers n and m : the number of vertices in the polygon and the number of points.

After this, there are n lines that describe the polygon. The i th such line has two integers x_i and y_i .

You may assume that the polygon is simple, i.e., it does not intersect itself.

Finally, there are m lines that describe the points. Each line has two integers x and y .

Output

For each point, print "INSIDE", "OUTSIDE" or "BOUNDARY".

Constraints

- $3 \leq n, m \leq 1000$
- $1 \leq m \leq 1000$
- $-10^9 \leq x_i, y_i \leq 10^9$
- $-10^9 \leq x, y \leq 10^9$

Example

Input:

```
4 3
1 1
4 2
3 5
1 4
2 3
3 1
1 3
```

Output:

```
INSIDE
OUTSIDE
BOUNDARY
```

CSES Problem Set

Polygon Lattice Points

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a polygon, your task is to calculate the number of lattice points inside the polygon and on its boundary. A lattice point is a point whose coordinates are integers.

The polygon consists of n vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The vertices (x_i, y_i) and (x_{i+1}, y_{i+1}) are adjacent for $i = 1, 2, \dots, n - 1$, and the vertices (x_1, y_1) and (x_n, y_n) are also adjacent.

Input

The first input line has an integer n : the number of vertices.

After this, there are n lines that describe the vertices. The i th such line has two integers x_i and y_i .

You may assume that the polygon is simple, i.e., it does not intersect itself.

Output

Print two integers: the number of lattice points inside the polygon and on its boundary.

Constraints

- $3 \leq n \leq 10^5$
- $-10^9 \leq x_i, y_i \leq 10^9$

Example

Input:

```
4
1 1
5 3
3 5
1 4
```

Output:

```
6 8
```

CSES Problem Set**Minimum Euclidean Distance**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a set of points in the two-dimensional plane, your task is to find the minimum Euclidean distance between two distinct points.

The Euclidean distance of points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

Input

The first input line has an integer n : the number of points.

After this, there are n lines that describe the points. Each line has two integers x and y . You may assume that each point is distinct.

Output

Print one integer: d^2 where d is the minimum Euclidean distance (this ensures that the result is an integer).

Constraints

- $2 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Example

Input:

```
4
2 1
4 4
1 2
6 3
```

Output:

```
2
```

CSES Problem Set

Convex Hull

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a set of n points in the two-dimensional plane, your task is to determine the convex hull of the points.

Input

The first input line has an integer n : the number of points.

After this, there are n lines that describe the points. Each line has two integers x and y : the coordinates of a point.

You may assume that each point is distinct, and the area of the hull is positive.

Output

First print an integer k : the number of points in the convex hull.

After this, print k lines that describe the points. You can print the points in any order. Print all points that lie on the convex hull.

Constraints

- $3 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Example

Input:

```
6
2 1
2 5
3 3
4 3
4 4
6 3
```

Output:

```
4
2 1
2 5
4 4
6 3
```

CSES Problem Set

Maximum Manhattan Distances

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A set is initially empty and n points are added to it. Calculate the maximum Manhattan distance of two points after each addition.

Input

The first line has an integer n : the number of points.

The following n lines describe the points. Each line has two integers x and y . You can assume that each point is distinct.

Output

After each addition, print the maximum distance.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Example

Input:

```
5
1 1
3 2
2 4
2 1
4 5
```

Output:

```
0
3
4
4
7
```

CSES Problem Set

All Manhattan Distances

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a set of points, calculate the sum of all Manhattan distances between two point pairs.

Input

The first line has an integer n : the number of points.

The following n lines describe the points. Each line has two integers x and y . You can assume that each point is distinct.

Output

Print the sum of all Manhattan distances.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Example

Input:

```
5
1 1
3 2
2 4
2 1
4 5
```

Output:

```
36
```

CSES Problem Set

Intersection Points

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given n horizontal and vertical line segments, your task is to calculate the number of their intersection points.

You can assume that no parallel line segments intersect, and no endpoint of a line segment is an intersection point.

Input

The first line has an integer n : the number of line segments.

Then there are n lines describing the line segments. Each line has four integers: x_1 , y_1 , x_2 and y_2 : a line segment begins at point (x_1, y_1) and ends at point (x_2, y_2) .

Output

Print the number of intersection points.

Constraints

- $1 \leq n \leq 10^5$
- $-10^6 \leq x_1 \leq x_2 \leq 10^6$
- $-10^6 \leq y_1 \leq y_2 \leq 10^6$
- $(x_1, y_1) \neq (x_2, y_2)$

Example

Input:

```
3
2 3 7 3
3 1 3 5
6 2 6 6
```

Output:

```
2
```

CSES Problem Set**Line Segments Trace I**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n line segments whose endpoints have integer coordinates. The left x-coordinate of each segment is 0 and the right x-coordinate is m . The slope of each segment is an integer.

For each x-coordinate $0, 1, \dots, m$, find the maximum point in any line segment.

Input

The first line has two integers n and m : the number of line segments and the maximum x-coordinate.

The next n lines describe the line segments. Each line has two integers y_1 and y_2 : there is a line segment between points $(0, y_1)$ and (m, y_2) .

Output

Print $m + 1$ integers: the maximum points for $x = 0, 1, \dots, m$.

Constraints

- $1 \leq n, m \leq 10^5$
- $0 \leq y_1, y_2 \leq 10^9$

Example

Input:

```
4 5
1 6
7 2
5 5
10 0
```

Output:

```
10 8 6 5 5 6
```

CSES Problem Set

Line Segments Trace II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n line segments whose endpoints have integer coordinates. Each x-coordinate is between 0 and m . The slope of each segment is an integer.

For each x-coordinate $0, 1, \dots, m$, find the maximum point in any line segment. If there is no segment at some point, the maximum is -1 .

Input

The first line has two integers n and m : the number of line segments and the maximum x-coordinate.

The next n lines describe the line segments. Each line has four integers x_1, y_1, x_2 and y_2 : there is a line segment between points (x_1, y_1) and (x_2, y_2) .

Output

Print $m + 1$ integers: the maximum points for $x = 0, 1, \dots, m$.

Constraints

- $1 \leq n, m \leq 10^5$
- $0 \leq x_1 < x_2 \leq m$
- $0 \leq y_1, y_2 \leq 10^9$

Example

Input:

```
4 5
1 1 3 3
1 2 4 2
2 4 5 7
2 8 5 2
```

Output:

```
-1 2 8 6 6 7
```

CSES Problem Set**Lines and Queries I**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to efficiently process the following types of queries:

1. Add a line $ax + b$
2. Find the maximum point in any line at position x

Input

The first line has an integer n : the number of queries.

The following n lines describe the queries. The format of each line is either "1 a b " or "2 x ".

You may assume that the first query is of type 1.

Output

Print the answer for each query of type 2.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq a, b \leq 10^9$
- $0 \leq x \leq 10^5$

Example

Input:

```
6
1 1 2
2 1
2 3
1 0 4
2 1
2 3
```

Output:

```
3
5
4
5
```

CSES Problem Set

Lines and Queries II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to efficiently process the following types of queries:

1. Add a line $ax + b$ that is active in range $[l, r]$
2. Find the maximum point in any active line at position x

Input

The first line has an integer n : the number of queries.

The following n lines describe the queries. The format of each line is either "1 a b l r " or "2 x ".

Output

Print the answer for each query of type 2. If no line is active, print NO.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq a, b \leq 10^9$
- $0 \leq x \leq 10^5$
- $0 \leq l \leq r \leq 10^5$

Example

Input:

```
6
1 1 2 1 3
2 3
2 4
1 0 4 1 5
2 3
2 4
```

Output:

```
5
NO
5
4
```

CSES Problem Set**Area of Rectangles**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given n rectangles, your task is to determine the total area of their union.

Input

The first line has an integer n : the number of rectangles.

After that, there are n lines describing the rectangles. Each line has four integers x_1 , y_1 , x_2 and y_2 : a rectangle begins at point (x_1, y_1) and ends at point (x_2, y_2) .

Output

Print the total area covered by the rectangles.

Constraints

- $1 \leq n \leq 10^5$
- $-10^6 \leq x_1 < x_2 \leq 10^6$
- $-10^6 \leq y_1 < y_2 \leq 10^6$

Example

Input:

```
3
1 3 4 5
3 1 7 4
5 3 8 6
```

Output:

```
24
```